

Manners Maketh MAN: Multi-Agent Norm Dynamics under Cultural Moral Values

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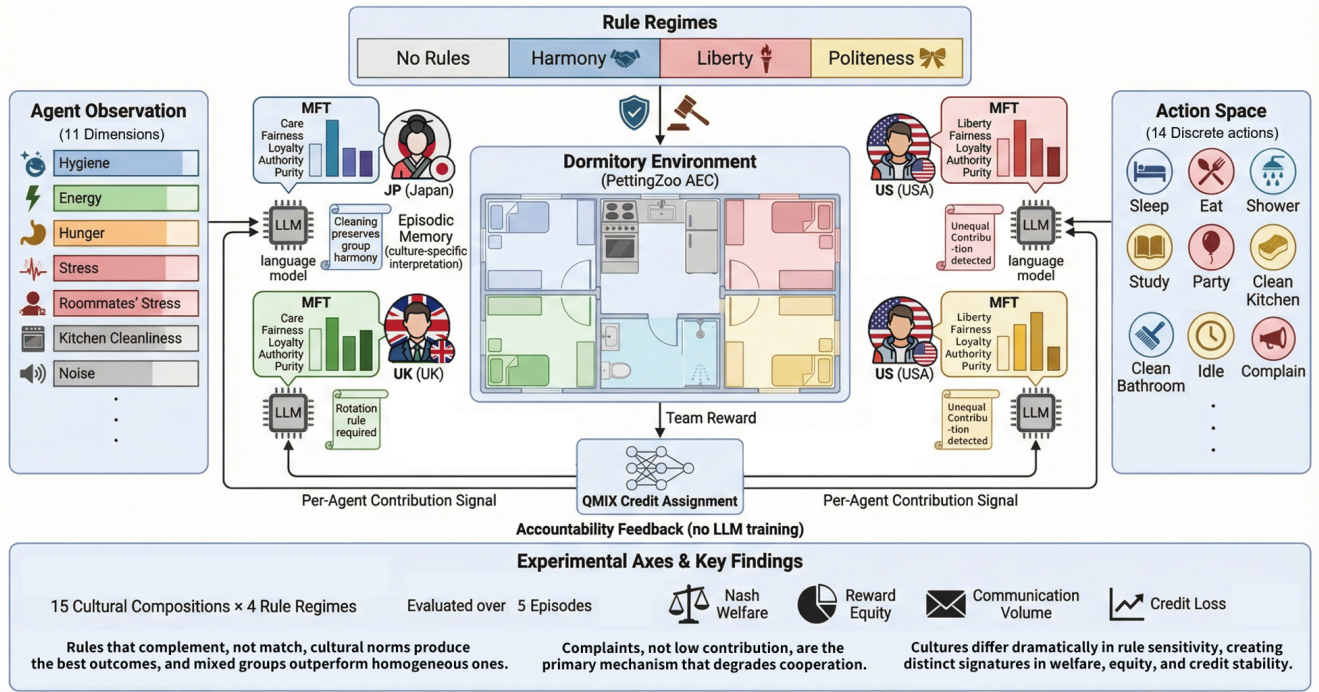


Figure 1: System architecture overview. Four LLM agents with culturally grounded personas (Asian, American, European), instantiated in our experiments using representative national profiles (Japan, the United States, and the United Kingdom), interact within a simulated co-living environment. The PettingZoo Agent Environment Cycle (AEC) provides 11-dimensional partial observations and 14 discrete actions. A frozen Large Language Model (LLM; GPT-4o-mini) generates decisions via a three-layer prompt (system/user/memory) parameterized by Moral Foundations Theory (MFT) profiles. A compressed episodic memory accumulates action history and accountability feedback. A QMIX-based credit assignment module decomposes team value into per-agent contribution signals via monotonic mixing. We evaluate 15 cultural compositions under 4 institutional rule regimes, with 5 episodes per condition, using collective welfare, reward equity, communication volume, and credit assignment stability metrics.

Abstract

When people from different cultures live together, conflicts often arise—not because of bad intentions, but because shared rules interact with different moral norms. Yet in real-world institutions, it is difficult to systematically test how population composition

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and rule design jointly shape collective outcomes. We study this question in a controlled setting using an LLM-based multi-agent dormitory simulator. Agents represent culture-conditioned moral preferences drawn from East Asian, North American, and European profiles, grounded in Moral Foundations Theory and equipped with episodic memory. The environment enforces explicit rule regimes (NoRules, Harmony, Liberty, Politeness) through both prompt-level constraints and environment-level penalties. To quantify the gap between individually driven behavior and collective welfare, we introduce a QMIX-based credit assignment module that produces per-agent contribution signals without updating LLM parameters. Across 15 cultural compositions and 4 rule regimes (60 conditions), we evaluate welfare, equity, communication, and credit stability. Contrary to the cultural fit hypothesis, we find that culturally mixed groups under complementary rules outperform homogeneous groups by up to 16.2%, and that rule sensitivity itself constitutes a distinct cultural signature.

CCS Concepts

- **Computing methodologies** → **Modeling and simulation**;
- **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms**.

Keywords

Moral Foundations Theory; Norm Dynamics; Multi-Agent Simulation; Large Language Models; Credit Assignment

1 Introduction

Shared social environments pose a recurring coordination problem. Individuals must coexist, share resources, and regulate everyday behavior under common rules [13]. Yet collective outcomes vary widely even when formal rules are explicit [20, 23]. A key reason is that people do not interpret acceptable behavior in the same way: conflicts over cleanliness, noise, or confrontation often reflect deeper moral intuitions about fairness, authority, and personal freedom [4, 10, 18]. These intuitions are culturally patterned and can shape how institutions function in practice [1, 4].

This issue is especially salient in *co-living spaces*—such as dormitories, shared apartments, and communal residences—where people interact repeatedly under partial observability and must balance informal norms with formal rules [5, 6]. Social theory has long argued that order depends not only on enforcement but also on internalized behavioral dispositions [7]. This motivates our central question: *how do population composition and institutional rule regimes jointly shape collective welfare and behavioral dynamics in shared environments?*

Empirically isolating this interaction in real-world settings is difficult because population composition, rule regimes, and enforcement conditions cannot usually be varied independently. We address this limitation with a controlled *Large Language Model* (LLM)-based multi-agent simulation grounded in *Moral Foundations Theory* (MFT) [9, 12]. We study three country-level persona instantiations derived from Japan, the United States, and the United Kingdom, and use the broader labels *Asian*, *American*, and *European* only as shorthand for these experimental anchors rather than as exhaustive regional categories. Agents are persona-prompted LLMs

with culture-conditioned moral priorities and compressed episodic memory.

Our environment models a university dormitory as a canonical cross-cultural co-living coordination problem. Agents share resources, interact repeatedly under partial observability, and act under one of four institutional rule regimes: NoRules, Harmony, Liberty, or Politeness. To connect individual behavior to collective outcomes, we incorporate a QMIX-based credit assignment module [22] that produces interpretable per-agent contribution signals without updating LLM parameters.

Our study is guided by three research questions:

- RQ1:** How do cultural composition and rule regimes jointly determine collective welfare in shared environments?
- RQ2:** Which behavioral and communication mechanisms explain differences across composition–rule configurations?
- RQ3:** How do rule regimes affect reward equity and the stability of credit assignment, that is, the learnability of cooperation?

We make three contributions. First, we introduce a controlled testbed for LLM-based social simulation that varies both population composition and institutional rule regimes. Second, we propose a minimal **ego-accountability** framework in which LLM agents generate behavior while QMIX provides interpretable per-agent contribution signals without LLM training. Third, across 15 cultural compositions and 4 rule regimes, we show that rules can either mitigate or amplify ego-optimum conflict, with corresponding differences in welfare, communication, and cooperative learnability.

2 Background and Related Work

2.1 Cultural Moral Psychology

Moral Foundations Theory (MFT) models moral judgment using a small set of foundations—Care, Fairness, Loyalty, Authority, and Purity, with Liberty proposed subsequently [9, 11, 12]. Cross-cultural studies show that these foundations are universal but systematically reweighted across societies: East Asian cultures emphasize Authority and Purity, whereas Western individualist cultures emphasize Care and Fairness [2, 9]. Such differences manifest in everyday norms governing cleanliness, conflict resolution, and courtesy [7]. While MFT is well established in psychology, it has rarely been used to parameterize agents in computational social simulation. Existing work on value alignment [8] and moral reasoning in LLMs typically treats values abstractly, rather than as empirically grounded, population-level moral priors interacting under shared institutions.

2.2 Social Norms and Collective Welfare

Social norms and their enforcement have been studied in game-theoretic, evolutionary, and multi-agent learning frameworks [3, 16, 19]. In multi-agent reinforcement learning (MARL), norm enforcement mechanisms such as punishment can improve collective outcomes [15]. However, these models usually assume homogeneous populations with shared interpretations of norms. Less is known about how explicit rules interact with heterogeneous moral dispositions, where the same behavior may be seen as compliant, insufficient, or provocative depending on cultural priors.

2.3 LLM-Based Social Simulation

LLMs have recently been used as agents in social simulations, enabling persistent memory, communication, and emergent coordination [14, 17, 21]. This line of work demonstrates the feasibility of language models as interactive social agents. Our work builds on this paradigm by grounding agent personas in empirical moral data and by varying explicit institutional rule regimes, using LLMs as a controlled testbed for exploring composition–institution interactions rather than for prediction.

2.4 Credit Assignment in Cooperative MARL

Credit assignment methods address how individual actions contribute to team-level outcomes. QMIX [22] decomposes global reward into per-agent value functions under a monotonicity constraint and is widely used in cooperative MARL. We use QMIX not for policy learning but as an accountability signal, allowing us to study how reward equity and credit stability vary across rule regimes and culturally heterogeneous populations.

3 Methodology

We study collective behavior under controlled variation of *population composition* and *institutional rule regimes* in a shared co-living environment. Figure 1 summarizes the full architecture: (i) a Petting-Zoo AEC environment providing partial observations and discrete actions, (ii) persona-driven LLM agents that operationalize culturally internalized moral dispositions, (iii) a compressed episodic memory system enabling in-context adaptation, and (iv) a QMIX-based credit assignment module providing post-hoc accountability signals without LLM training.

3.1 Environment: co-living dynamics and rule regimes

We implement a shared co-living space as a PettingZoo AEC environment with $N = 4$ agents, partial observability, and discrete-time dynamics. At each timestep, every agent receives an **11-dimensional** observation vector capturing both individual physiological state and shared environmental state (e.g., hygiene, energy, hunger, stress, kitchen cleanliness, bathroom occupancy, noise, time of day, and roommates’ stress), and selects one of **14** discrete actions (e.g., sleep/eat/shower/study, cleaning actions, and communication actions such as complaints or indirect hints). Full definitions of the observation and action spaces are provided in Appendix A.

Institutional design is represented by a *global rule regime* shared across all agents. We consider four regimes: NoRules (baseline), HARMONY, LIBERTY, and POLITENESS, each specifying permissible behavior and inducing compliance costs upon violations. These regimes are intentionally defined at the environment level to isolate the causal effect of *external rules* from agents’ internal preferences. Complete rule specifications are provided in Appendix C.

3.2 Agents (Ego): culturally grounded LLM personas

Each agent is a *persona-prompted* Large Language Model (LLM) that generates actions and optional messages from its current observation and memory context, while keeping model parameters

frozen. Agents are grounded in Moral Foundations Theory (MFT) [2, 9, 12] through culture-conditioned moral priority profiles over Care, Fairness, Loyalty, Authority, and Purity.

To avoid over-claiming cultural universality while retaining controlled variation, we report cultural groups at a **regional level** in the main text (**Asian, American, European**) and **instantiate** them in experiments using representative country-level survey profiles (Japan/US/UK). We use these instances as operational anchors rather than exhaustive cultural categories; all raw MFT scores used for persona construction are listed in Appendix B, along with example persona prompts in Appendix H.

3.3 Prompting and episodic memory: in-context adaptation

As shown in Figure 1, each agent acts via a three-layer prompt: (i) a **system prompt** specifying cultural persona, MFT profile, and current rule regime, (ii) a **user prompt** containing the current observation, and (iii) a **memory prompt** containing a compressed episodic summary of recent experiences. Agents output a structured response (JSON) with an action choice, brief reasoning, and an optional message to roommates.

Episodic memory is designed to support adaptation *in context* rather than through parameter updates: it stores bounded-length summaries of action–outcome patterns, key lessons, recent action–reward tuples, and observations about others. The memory representation is kept compact (bounded prompt length) to ensure consistent inference-time behavior across conditions. Full prompt and memory formats are provided in Appendix E.

3.4 Accountability: QMIX-based credit assignment without LLM training

To connect ego-driven behavior to collective outcomes, we introduce an external accountability channel via QMIX [22]. QMIX decomposes the team-level value into **per-agent contribution signals** through a monotonic mixing function. Importantly, *only* QMIX parameters are trained; the LLM agents remain frozen. The resulting per-agent contribution signals are used as interpretable accountability feedback that is appended to each agent’s episodic memory, enabling analysis of reward equity, conflict escalation, and the stability of cooperative learning across composition–rule combinations. The full QMIX formulation, training details, and reward definition are provided in Appendix F.

3.5 Experimental design and evaluation metrics

We evaluate **15** population compositions (homogeneous and mixed groups across Asian/American/European personas) under **4** rule regimes, yielding **60** conditions in total. Each condition is run for **5** episodes of **100** timesteps. The complete list of compositions, hyperparameters, and runtime details is provided in Appendix D.

Our primary outcomes include: (i) collective welfare measured via Nash Social Welfare, (ii) reward equity derived from per-agent contribution signals, (iii) communication volume (messages exchanged), and (iv) credit stability captured by QMIX learning signals (e.g., credit loss). Complete results tables, hypothesis checks, and violation statistics are reported in Appendix G and Appendix I.

Table 1: Mean Nash Social Welfare for all 60 experimental conditions (15 compositions $\times$ 4 rule regimes, 5 episodes each). Bold indicates the best-performing rule for each composition.

Composition	NoRules	Liberty	Harmony	Politeness
JP4	8.35	12.36	9.74	11.81
US4	10.61	11.73	11.13	11.51
UK4	10.65	8.74	14.91	6.80
JP3_US1	11.69	8.95	12.97	12.50
JP3_UK1	10.51	8.84	8.85	8.93
US3_JP1	8.95	9.93	15.70	10.90
US3_UK1	9.62	11.47	11.22	12.47
UK3_JP1	9.18	8.47	6.31	10.39
UK3_US1	17.33	10.22	12.45	14.23
JP2_US2	8.63	5.22	16.82	12.56
JP2_UK2	6.59	8.05	9.46	9.85
US2_UK2	9.21	16.72	16.58	9.87
JP2_US1_UK1	12.20	7.91	15.16	10.60
US2_JP1_UK1	7.89	16.10	12.52	14.77
UK2_JP1_US1	9.92	8.11	11.52	13.47

4 Results

4.1 RQ1: Cultural Composition, Rule Regimes, and Collective Welfare

Table 1 reports mean Nash Social Welfare for all 60 conditions. Three core patterns emerge.

Mixed groups outperform homogeneous groups. The top performer is UK3_US1 + NoRules (17.33), exceeding the best homogeneous condition UK4 + Harmony (14.91) by 16.2%. Nine of the top ten conditions are mixed compositions.

Cultural fit hypothesis rejected. Agents perform better under rules that *complement* rather than *match* their cultural background. JP4 achieves highest welfare under Liberty (12.36), not Harmony (9.74); UK4 thrives under Harmony (14.91), not Politeness (6.80). Only US4 matches the expected pattern (Liberty = 11.73). Rule sensitivity itself varies by culture: UK4 shows extreme sensitivity (gap = 8.11 between best and worst rule), JP4 moderate (gap = 4.01), and US4 minimal (gap = 1.12).

Harmony rules dominate overall. Across all compositions, Harmony achieves the highest welfare in 6 of 15 compositions, followed by Politeness (4), Liberty (3), and NoRules (2)—though this aggregate masks strong composition-dependent variation.

Answer to RQ1: Rule regime interacts non-trivially with cultural composition. The cultural fit hypothesis is rejected in favor of cultural complementarity, mixed groups outperform homogeneous ones, and rule sensitivity varies dramatically by cultural profile.

4.2 RQ2: Behavioral Mechanisms

The dominant behavioral differentiator is complaint frequency (Table 2). US agents complain at 19.8% of timesteps, compared with 0% for JP and 0.4% for UK agents. This Fairness-monitoring behavior

Table 2: Behavioral patterns for homogeneous groups (% of actions).

	Complain	Hint	Avoid	Clean	Study
JP4	0.0	0.9	12.3	10.5	28.6
US4	19.8	1.3	9.0	17.7	12.9
UK4	0.4	4.9	6.5	14.2	25.9

Table 3: QMIX credit loss and reward variance by composition.

	NoRules	Liberty	Harmony	Polit.	Reward Var
JP4	4.90	3.08	2.60	3.03	4,133
US4	1.80	1.92	5.13	5.14	11,998
UK4	1.20	2.22	3.05	4.81	4,235

peaks under Liberty rules (28.2%), suggesting that direct communication norms license grievance escalation rather than resolve conflict.

JP and UK agents achieve comparable welfare through distinct non-confrontational strategies: JP agents *avoid* (12.3%) and maintain routines (Study 28.6%); UK agents deploy *hints* (4.9%) for indirect coordination. US agents clean more (17.7% vs 10.5%) yet achieve worse welfare—reactive compensation cannot offset complaint-driven conflict.

Message volume correlates negatively with welfare: JP4 sends 2,805 messages (fewest), US4 sends 4,122 (most). The difference lies in content: JP declarations (“I’ll clean”) versus US demands (“Someone needs to step up”).

Answer to RQ2: Complaint behavior, not contribution rate, drives welfare differences. High-Fairness monitoring produces escalation; avoidance and hinting prevent it.

4.3 RQ3: Credit Assignment and Reward Equity

Credit stability depends on composition–rule interaction (Table 3). US4 shows a striking split: stable under NoRules/Liberty (1.80–1.92) but destabilized under Harmony/Politeness (5.13–5.14), suggesting explicit prosocial rules create tension with individualist orientations. UK4 degrades progressively from NoRules (1.20) to Politeness (4.81)—the culturally “matched” regime produces worst stability.

Reward variance reveals inequity: US4 exhibits 3 $\times$ higher inter-agent variance (11,998) than JP4 (4,133) or UK4 (4,235). Fairness-driven blame concentrates negative outcomes on specific agents, while JP4’s implicit coordination distributes rewards more equitably.

Answer to RQ3: Credit stability depends on composition–rule fit, not composition alone. Complaint-driven groups show higher reward inequity and less stable credit learning.

5 Discussion

5.1 Cultural Complementarity vs. Cultural Fit

The results consistently reject the cultural fit hypothesis that groups perform best under rules matching their cultural background. Instead, outcomes follow a principle of **cultural complementarity**:

external rules are most effective when they add constraints that are not already internalized by agents. Rules that mirror existing moral dispositions tend to be redundant or over-constraining, while rules that provide complementary structure improve coordination. This pattern holds across homogeneous and mixed groups and indicates that institutional design should focus on how rules interact with internal moral priorities, rather than on cultural labels alone.

5.2 Rule Sensitivity as a Cultural Signature

Groups differ markedly in how sensitive they are to changes in rule regimes. Some populations show large performance swings depending on which rules are imposed, while others behave similarly across regimes. This variation reflects differences in the strength of coupling between internalized norms and external institutions. Rule sensitivity therefore serves as a diagnostic signal: in highly sensitive populations, policy choice is critical, whereas in less sensitive populations, rule changes mainly affect surface behavior rather than collective outcomes.

5.3 Communication as a System Load

Higher communication volume is associated with poorer collective outcomes, suggesting that communication often reflects coordination failure rather than enabling coordination. In particular, complaint-based communication correlates with instability, as it triggers defensive and reactive behavior. By contrast, indirect or implicit coordination strategies—such as routine maintenance or non-confrontational signaling—support more stable interaction. This implies that the effectiveness of communication depends less on quantity than on its functional role within the interaction dynamics.

5.4 Fairness and Endogenous Failure Modes

The findings reveal an endogenous failure mode of fairness-oriented behavior. When fairness operates primarily as monitoring and attribution, early imbalances can escalate into blame dynamics through memory and communication, increasing inequity without any explicit punishment mechanism. In such cases, accountability mechanisms destabilize cooperation rather than support it. This highlights a tension between fairness as a moral ideal and fairness as an operational principle in shared systems.

5.5 Violations and Active Engagement

Norm violations are not uniformly detrimental to collective performance. In several conditions, higher welfare coincides with more frequent violations, indicating active engagement rather than disengagement. Strict compliance can reflect withdrawal from interaction, while occasional norm overstepping may signal participation and mutual adjustment. This distinction suggests that compliance alone is an insufficient proxy for cooperation and must be interpreted alongside interaction patterns and outcomes.

5.6 Implications for Policy Design under Heterogeneous Moral Priorities

Taken together, the results suggest that effective policy design in heterogeneous populations cannot rely on prescribing a single

culture-specific set of manners. Instead, policies should be selected based on how they complement the moral priorities present in the population and how they shape communication and escalation pathways. Mixed populations often benefit from such complementarity, as different moral orientations offset one another under appropriately chosen rules. The broader contribution of this work is to demonstrate how controlled simulation can make policy–norm interactions explicit, supporting comparative reasoning about which institutional designs are likely to support cooperation under different moral configurations.

6 Conclusion

This study examines how collective outcomes in shared environments emerge from the interaction between internalized moral priorities and external institutional rules. Using an LLM-based multi-agent simulation with culturally grounded personas, episodic memory, and explicit credit assignment, we varied population composition and rule regimes to analyze their effects on cooperation, equity, and stability.

Three core insights emerge. First, the cultural fit hypothesis is rejected: groups do not reliably perform best under rules that mirror their cultural background. Instead, outcomes depend on *complementarity* between internalized norms and external constraints. Second, heterogeneous populations often outperform homogeneous ones when moral priorities complement one another and rules avoid escalation pathways, suggesting that diversity can support coordination rather than friction. Third, populations differ in rule sensitivity, indicating that rule choice should be treated as population-dependent rather than universal.

The study has clear limitations. Cultural personas are instantiated from population-level averages, abstracting away within-group heterogeneity and individual variation. The number of episodes per condition is limited, constraining statistical power for fine-grained effects. Agents rely on a single LLM with bounded episodic memory, and interactions are restricted to small group sizes. Future work should extend this framework by modeling mixed moral profiles, increasing interaction scale, evaluating multiple model families, and introducing endogenous norm formation.

Overall, this work does not propose normative policies. Rather, it shows how controlled simulation can probe norm–institution interactions that are difficult to isolate in the real world and can support comparative reasoning about which rule regimes are more likely to sustain cooperation in heterogeneous populations.

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A Co-living Simulation Environment (PettingZoo AEC)

We implement the co-living environment as a PettingZoo AEC environment with $N = 4$ agents, partial observability, and discrete-time dynamics. Each episode consists of 100 timesteps (approximately four simulated days at hourly resolution).

A.1 Observation Space

Each agent receives an 11-dimensional observation vector:

$$\text{Obs}_i = [C_k, B, N, t, h_i, e_i, g_i, s_i, s_{-i}^{(1)}, s_{-i}^{(2)}, s_{-i}^{(3)}],$$

where C_k denotes kitchen cleanliness, B bathroom occupancy, N noise level, t time of day, (h_i, e_i, g_i, s_i) the agent’s hygiene, energy, hunger, and stress, and s_{-i} the stress levels of roommates.

A.2 Action Space

Agents select from fourteen discrete actions: Sleep, Eat, Shower, Cook, Study, Party, Invite_Guest, Clean_Kitchen, Clean_Bathroom, Complain, Hint, Avoid, Borrow_Item, and Idle.

B Cultural Personas and Moral Foundations

Agent personas are parameterized by Moral Foundations Theory (MFT) scores derived from cross-national survey data [2, 9]. In the main text we refer to regional groupings (**Asian**, **American**, **European**), and instantiate them with representative country-level profiles (Japan/US/UK) as operational anchors for controlled experiments.

Table 4: Raw MFT scores used for agent personas (country-level instances).

	Care	Fairness	Loyalty	Authority	Purity
Japan	22.74	18.57	14.36	15.95	16.60
US	22.00	22.00	15.00	15.00	12.00
UK	21.00	20.00	16.00	15.00	14.00

Although Care scores are similar across cultures, their behavioral expression differs. Identical experiences can be interpreted through different moral lenses: Japan agents may frame cleaning as preserving group harmony (Care, Purity), US agents may detect contribution imbalances (Fairness), and UK agents may propose procedural solutions.

C Social Norm Rule Sets

We define four rule regimes at the environment level. In the main text we discuss them as institutional styles (HARMONY, LIBERTY, POLITENESS) that align with broader cultural logics, while in the implementation we operationalize them using concrete rule lists inspired by Japan/US/UK norms to ensure specificity and reproducibility.

C.1 NoRules (Baseline)

No explicit social norms enforced. Agents act based solely on cultural personas.

C.2 Harmony Rules (Japan-inspired operationalization)

- (1) Quiet hours after 10 PM
- (2) Check availability before using shared spaces
- (3) Clean immediately after use
- (4) Use hints instead of complaints
- (5) Prioritize group harmony

C.3 Liberty Rules (US-inspired operationalization)

- (1) Full freedom in private spaces
- (2) Respond only to explicit requests
- (3) Direct complaints encouraged
- (4) First-come, first-served resources
- (5) Personal responsibility

C.4 Politeness Rules (UK-inspired operationalization)

- (1) Indirect, softened expression
- (2) Queue respect
- (3) Prefer avoidance over confrontation
- (4) Apologetic framing
- (5) Privacy respect

D Experimental Configuration

Table 5: Cultural compositions (15 conditions).

Type	Composition	JP	US	UK
Homogeneous	JP4, US4, UK4	4/0/0	0/4/0	0/0/4
3:1 Mixed	JP3_US1, JP3_UK1, ...	6 conditions		
2:2 Mixed	JP2_US2, JP2_UK2, US2_UK2	3 conditions		
2:1:1 Mixed	JP2_US1_UK1, ...	3 conditions		

Mapping to main-text labels. In the main text we refer to JP/US/UK-based personas as Asian/American/European respectively, while retaining JP/US/UK notation here for compact experimental book-keeping and reproducibility.

Table 6: Experimental hyperparameters.

Parameter	Value
Total conditions	60 (15 compositions $\times$ 4 rules)
Episodes per condition	5
Timesteps per episode	100
LLM model	GPT-4o-mini
Total runtime	170.8 minutes

E Agent Architecture

E.1 Prompt Structure

Each agent receives a three-layer prompt: **System prompt** (cultural identity, MFT weights, rule regime), **User prompt** (current

observation), **Memory prompt** (compressed episodic memory). Agents respond with JSON containing reasoning, action, and optional message.

E.2 Episodic Memory

Each agent maintains: (1) aggregate statistics, (2) lessons learned (up to 5), (3) recent action–reward tuples, (4) observations of others. Prompt length bounded to ~700 tokens.

F QMIX and Reward Structure

F.1 Credit Assignment

QMIX decomposes team reward into per-agent contributions: $Q_{\text{tot}} = f_{\text{mix}}(Q_1, \dots, Q_4; s)$. Only QMIX parameters are trained; LLMs remain frozen.

Table 7: QMIX hyperparameters.

Hidden layers	(256, 128)
Mixing hidden dim	64
Learning rate	1×10^{-4}
Discount (γ)	0.99

F.2 Reward Structure

$$R_i(t) = R_{\text{bio}}^i + R_{\text{moral}}^i - C_{\text{compliance}}^i \quad (1)$$

where R_{bio} captures physiological needs (hygiene, energy, hunger, stress), R_{moral} is MFT-weighted, and

$$C_{\text{compliance}} = 0.5 \cdot \#[\text{violation}]. \quad (2)$$

Nash Social Welfare:

$$W = \left(\prod_{i=1}^4 \max(R_i, \epsilon) \right)^{1/4} \quad (3)$$

G Complete Experimental Results

Table 8: Homogeneous group results. Bold = best rule.

Comp	Rule	Nash	Std	Viol	Msgs
JP4	NoRules	8.35	2.46	0	821
	Liberty	12.36	2.81	32	558
	Harmony	9.74	3.58	131	708
	Politeness	11.81	3.99	49	718
US4	NoRules	10.61	3.19	0	1044
	Liberty	11.73	2.53	42	1064
	Harmony	11.13	3.20	86	1050
	Politeness	11.51	5.05	80	964
UK4	NoRules	10.65	2.97	0	1065
	Liberty	8.74	2.17	14	950
	Harmony	14.91	4.35	102	970
	Politeness	6.80	2.89	67	935

Table 9: Cultural fit hypothesis test.

Culture	Expected	W	Actual Best	W
JP4	Harmony	9.74	Liberty	12.36
US4	Liberty	11.73	Liberty	11.73
UK4	Politeness	6.80	Harmony	14.91

H Agent Persona Prompts

H.1 Japanese Agent Example

You are Yuki, a 22-year-old Japanese exchange student studying Computer Science.

Cultural Background:

- Value harmony (wa) and group cohesion above individual desires
- Prefer indirect communication to avoid confrontation
- High sensitivity to others' emotions and unspoken needs
- Respect for shared spaces and quiet hours
- High-context communication style

Moral Foundations (scaled 0-100):

- Care: 76 (protect others from harm)
- Fairness: 62 (reciprocity, equal treatment)
- Loyalty: 48 (group solidarity)
- Authority: 53 (respect hierarchy)
- Purity: 55 (avoid contamination)

Communication Style:

- Use hints and suggestions rather than direct requests
- Apologize preemptively even when not at fault
- Read the air (kuuki wo yomu) before acting

H.2 American Agent Example

You are Jake, a 21-year-old American student studying Business Administration.

Cultural Background:

- Value individual freedom and self-expression
- Prefer direct, explicit communication
- Comfortable with friendly confrontation
- Expect personal space and autonomy
- Low-context communication style

Moral Foundations (scaled 0-100):

- Care: 73 (protect others from harm)
- Fairness: 73 (equal rights, justice)
- Loyalty: 50 (patriotism, team spirit)
- Authority: 50 (respect for leadership)

- Purity: 40 (personal choice matters)

Communication Style:

- Say what you mean directly
- Ask explicitly for what you need
- Friendly debate is healthy

H.3 British Agent Example

You are Oliver, a 22-year-old British student studying History.

Cultural Background:

- Value politeness and proper etiquette
- Master of understatement and irony
- Queue respect and procedural fairness
- Privacy conscious, avoid intrusion
- Reserved but not unfriendly

Moral Foundations (scaled 0-100):

- Care: 70 (concern for welfare)
- Fairness: 67 (procedural justice)
- Loyalty: 53 (tradition, institutions)
- Authority: 50 (subtle hierarchy)
- Purity: 47 (propriety)

Communication Style:

- Use softening phrases ("I'm terribly sorry but ...")
- Understate problems ("slight issue" = major problem)
- Avoid direct confrontation; prefer hints

Harmony rules generate the most violations (mean = 105.5) and also achieve the highest welfare in several compositions (6 of 15), though this relationship is not universal—e.g., JP4 shows high violations under Harmony (131) without corresponding welfare gains. This suggests that violations may reflect active engagement in some configurations, but the violation–welfare relationship is moderated by cultural composition.

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I Violations by Rule Set

Table 10: Total rule violations across 5 episodes for each condition.

Composition	NoRules	Liberty	Harmony	Politeness
JP4	0	32	131	49
US4	0	42	86	80
UK4	0	14	102	67
JP3_US1	0	40	131	82
JP3_UK1	0	34	90	69
US3_JP1	0	42	80	78
US3_UK1	0	41	108	76
UK3_JP1	0	16	90	68
UK3_US1	0	23	125	77
JP2_US2	0	54	110	77
JP2_UK2	0	22	104	76
US2_UK2	0	36	88	67
JP2_US1_UK1	0	38	110	67
US2_JP1_UK1	0	43	90	66
UK2_JP1_US1	0	40	137	53
Mean	0	34.5	105.5	70.1